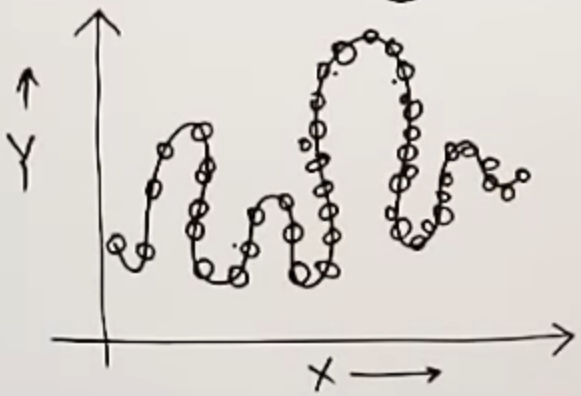
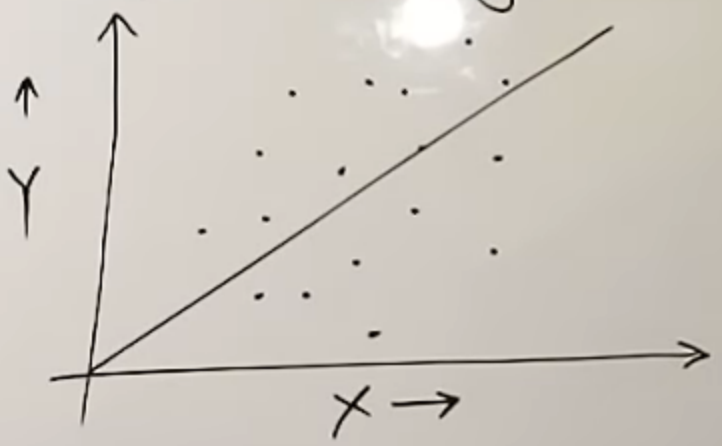
Overfitting and Underfitting





- Sphere
- Play
- Eat
- Radius = 5cm





→ Ball

Sphere



What is Overfitting

Overfitting occurs when a machine learning model learns the training data too well, capturing noise or random fluctuations in the data that don't represent the underlying patterns. As a result, an overfitted model performs well on the training set but fails to generalize to new, unseen data.

signs that your model might be overfitting:

1. **High Training Accuracy, Low Validation Accuracy:** The model achieves high accuracy on the training data but performs poorly on a separate validation set.
2. **Model Complexity:** Overfit models tend to be excessively complex, capturing noise in the training data instead of the underlying patterns.
3. **Large Differences Between Training and Validation Performance**: If there is a significant gap between the training and validation performance metrics, it may indicate overfitting.

Lert's see how we can solve it

Overfitting, underfitting in Machine Learning

Overfitting and **underfitting** are common challenges in machine learning that arise during the training of models.

Overfitting

Definition:

Overfitting occurs when a machine learning model learns not only the underlying patterns in the training data but also captures noise, outliers, and random fluctuations that are specific to the training set.

Characteristics:

- *The model performs extremely well on the training data but poorly on new, unseen data.*
- *It may capture the noise or peculiarities of the training data rather than the underlying patterns.*

Overfitting

Causes:

- *Too complex model: Models with too many parameters or high complexity may fit the training data too closely.*
- *Insufficient data: When there is not enough diverse data to represent all possible scenarios, the model may overfit to the limited examples available.*

Prevention and Remedies

- *Use simpler models.*
- *Regularization techniques: Add penalties to the model for complexity.*
- *Cross-validation: Split the data into multiple subsets for training and validation to assess generalization performance.*

Underfitting

Definition:

Underfitting occurs when a machine learning model is too simple to capture the underlying patterns in the training data. It fails to learn the complexities and nuances of the data.

Characteristics

- *The model performs poorly on both the training data and new, unseen data.*
- *It fails to capture important patterns and trends in the data.*

Underfitting

Causes:

- *Too simple model: Models with too few parameters or low complexity may not have the capacity to capture complex relationships.*
- *Insufficient training: If the model is not trained for a sufficient number of iterations, it may not converge to an optimal solution.*

Prevention and Remedies:

- *Use more complex models.*
- *Increase the number of training iterations.*
- *Ensure sufficient data is available for the model to learn meaningful patterns.*

Balancing Overfitting and Underfitting

- **Regularization:**
  - *Techniques such as L1 or L2 regularization add penalties to the model's parameters, discouraging excessive complexity.*
- **Cross-Validation:**
  - *Use techniques like k-fold cross-validation to assess the model's performance on multiple subsets of the data.*
- **Feature Engineering:**
  - *Carefully select and preprocess features to provide the model with relevant information.*
- **Ensemble Methods:**
  - *Combining multiple models can sometimes help mitigate overfitting and underfitting.*
- **Early Stopping:**
  - *Monitor the model's performance on a validation set during training and stop when the performance starts degrading.*